Annexure3b- Complete filing

INVENTION DISCLOSURE FORM

Details of Invention for better understanding:

1. TITLE: AI-Based Safety Gear Detection and Traffic Signal Control System

2. INTERNAL INVENTOR(S)/ STUDENT(S): All fields in this column are mandatory to be filled

US6989766B2	The invention relates to a traffic signal control system for controlling a plurality of signal junctions comprising a signal group oriented multi-agent control scheme, each agent operates independently and represents one or more traffic signals at a signal junction; means for each agent for determining traffic conditions at its signal junction and traffic conditions at neighbouring agents; and means for applying fuzzy logic in signal control operations, wherein signal control operation is based on traffic conditions at each agent and one or more neighbouring agents, such that the control operation is distributed to each agent to control each of said plurality of signal junctions.
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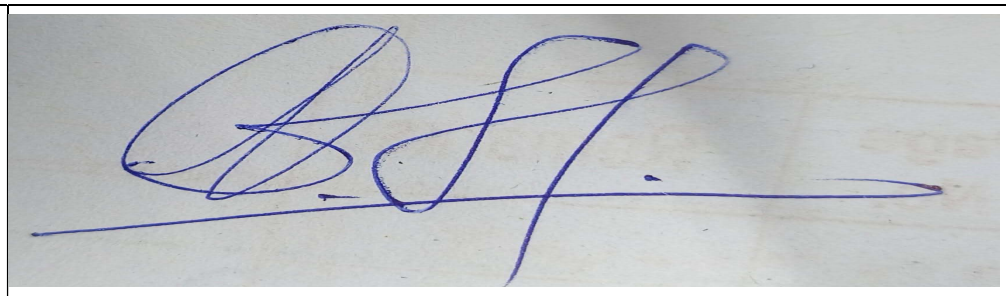
Signature (Mandatory)	
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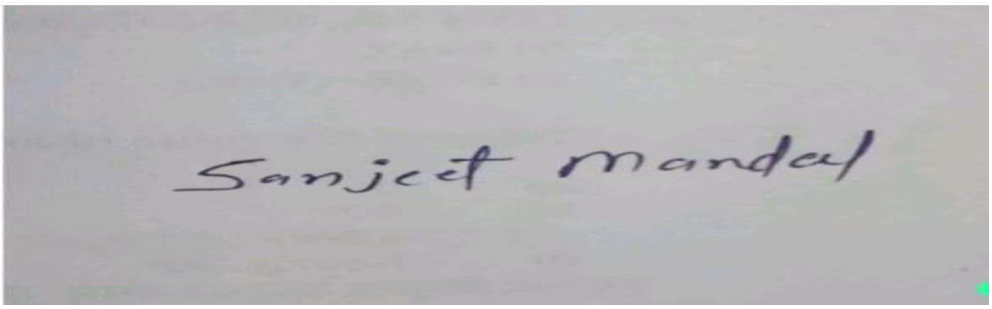
Signature (Mandatory)	
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3. DESCRIPTION OF THE INVENTION:

The present invention relates to an innovative system designed to enhance road safety by ensuring compliance with safety gear usage (such as seatbelts and helmets) before granting drivers and passengers access to move through traffic signals. More specifically, the invention leverages artificial intelligence (AI) and image recognition technology to detect whether a driver and/or passenger are wearing the appropriate safety gear, and uses this information to control the activation of traffic lights.

In today's world, many individuals fail to wear seatbelts or helmets while driving or riding, despite the established safety benefits. The invention addresses this concern by integrating AI-based detection systems into existing traffic signal infrastructure. The system uses AI algorithms to identify whether the required safety gear is in place, such as checking if the seatbelt is properly fastened or if a helmet is worn by the rider.

Upon detecting that the necessary safety gear is not being worn, the system delays or prevents the activation of the traffic light, ensuring that the vehicle does not proceed until the safety gear is properly worn. Once the system verifies that all required gear is in place, the traffic signal will change to green, allowing the vehicle to proceed.

This system can be implemented with the use of cameras or sensors embedded within the traffic light infrastructure. It may include real-time image recognition or other sensor-based technologies that provide accurate monitoring of vehicle occupants.

Key Benefits of the Invention:

1. Improved Road Safety: By enforcing seatbelt and helmet usage, the system significantly reduces the risk of injury and fatalities in traffic accidents.

2. Automation and Real-Time Enforcement: The system operates automatically, without the need for human intervention, and can be seamlessly integrated into existing traffic control systems.

3. Encouragement of Compliance: The use of traffic signal control as a deterrent encourages drivers and passengers to adhere to safety regulations more consistently.

4. Cost-Effective: The technology can be incorporated into existing infrastructure, making it an affordable solution for cities and regions looking to improve road safety.

5. The invention has the potential to be used worldwide as a preventive safety measure, improving overall traffic safety and ensuring that individuals are using the proper safety equipment when on the road.

A. PROBLEM ADDRESSED BY THE INVENTION:

1. **Non-Compliance with Safety Regulations**:
    - Despite the well-known safety benefits of wearing seatbelts and helmets, a significant number of drivers and passengers fail to use them regularly, often disregarding road safety laws.
    - This non-compliance increases the risk of serious injuries and fatalities in road accidents, especially in the event of a collision or sudden braking.

2. **Inconsistent Enforcement of Safety Gear Usage**:
    - Current methods of ensuring seatbelt and helmet use mainly rely on law enforcement officers performing manual checks or issuing fines. This can lead to inconsistent enforcement.
    - In busy traffic areas or during peak hours, it's difficult for law enforcement to monitor and enforce seatbelt and helmet use effectively.
    - Many drivers may ignore safety gear regulations, believing there is little chance of being caught.

3. **Lack of Immediate Deterrence**:
    - There is currently no real-time mechanism that can automatically prevent individuals from proceeding if they are not wearing their seatbelt or helmet.
    - Without an immediate deterrent, some drivers continue to take unnecessary risks, putting both themselves and others on the road at risk.

4. **Traffic Congestion and Safety:**
    - In cities with heavy traffic, traffic signal systems typically focus on managing flow, without addressing the critical issue of ensuring safety gear usage.

- Traditional systems do not account for individual safety compliance, leaving a gap in the overall goal of road safety.

5. **Public Awareness and Behaviour Change**:
    - Many people are unaware of the long-term consequences of not wearing a seatbelt or helmet, and even when they are aware, they may not view non-compliance as a significant threat.
    - There is a need for a system that not only monitors compliance but actively encourages individuals to wear the required safety gear.

B. OBJECTIVE OF THE INVENTION (Provide minimum two)

1. **To Automate the Enforcement of Seatbelt and Helmet Usage**:
    - The primary objective of this invention is to develop an AI-driven system that automatically detects whether vehicle occupants are wearing their seatbelts or helmets. This system enforces compliance by controlling the traffic signal, ensuring that vehicles cannot proceed unless the safety gear is properly worn. This reduces the reliance on manual checks by law enforcement and ensures consistent enforcement across all drivers and passengers.

2. **To Improve Road Safety by Reducing Accidents and Fatalities**:
    - Another key objective is to enhance road safety by promoting the use of seatbelts and helmets. By preventing vehicles from moving through traffic signals without the required safety gear, this system directly addresses the high number of road injuries and fatalities resulting from non-compliance with safety regulations. The invention aims to reduce these risks by making safety gear usage mandatory before allowing vehicles to proceed.

C. STATE OF THE ART/ RESEARCH GAP/NOVELTY:

State of the Art:

Currently, there are various technologies and systems aimed at enforcing road safety regulations, particularly concerning seatbelt and helmet use. However, most of these solutions are either manual, inconsistent, or lack real-time enforcement. Existing systems, such as traffic cameras or law enforcement patrols, are typically focused on detecting violations after they occur (e.g., through fines or tickets). These methods have limitations in terms of immediacy and the ability to prevent non-compliance before it happens.

Some research has explored the use of AI and computer vision for detecting road violations, but these technologies often focus on identifying other types of infractions, such as speeding or red-light violations. While there have been attempts to monitor seatbelt usage through sensors, these systems are limited in scope and typically do not incorporate automated traffic control or real-time enforcement.

Research Gap:

There is a significant gap in solutions that integrate real-time AI-based detection of seatbelt and helmet usage with automated traffic signal control. Current systems lack an immediate, automated deterrent that prevents vehicles from moving through traffic signals unless safety gear is worn, making them less effective in promoting consistent safety compliance. The research gap includes:

1. Real-Time Enforcement: While there are systems that detect seatbelt use, there is no widespread system that integrates this detection with real-time traffic signal management to prevent vehicles from proceeding if the safety gear is not worn.

2. Automated Traffic Signal Integration: Existing AI technologies for traffic control do not typically consider individual safety compliance as a factor in traffic light management.

3. Cost-Effective and Scalable Solutions: Many existing technologies either require expensive, complex infrastructure or are not easily scalable across large urban areas.

4. Comprehensive Safety Gear Detection: Most current systems either focus on seatbelt compliance or helmet usage separately. An integrated system that monitors both seatbelt and helmet compliance across different types of vehicles is not common.

Novelty of the Invention:

The present invention addresses these gaps by integrating AI-based detection systems with existing traffic signal infrastructure, creating a unique and novel solution. The key aspects that make this invention novel are:

1. **Dual Safety Gear Detection:**

    - The invention uniquely combines the detection of both seatbelt and helmet usage for all types of vehicles, including cars and motorcycles. The AI system **uses real**time image recognition or sensor data to check if both safety gear requirements are being met before allowing vehicles to proceed through traffic signals.

2. **AI-Controlled Traffic Signal Integration:**

    - Unlike existing technologies that simply monitor safety compliance, this system automatically links the detection of seatbelt and helmet usage to the control of traffic signals. Vehicles will not be allowed to pass through the signal unless the safety gear is properly worn, providing a real-time, automated solution for traffic safety enforcement.

3. **Seamless and Scalable Implementation:**

    - The invention is designed to be easily integrated with existing traffic control systems, making it cost-effective and scalable. Cities can implement the system without the need for major overhauls to their infrastructure, ensuring that it can be adopted widely.

4. **Real-Time Compliance and Enforcement:**

    - The real-time nature of the system means that compliance is verified immediately, providing a preventative measure rather than just a reactive one. This ensures that individuals who fail to wear their safety gear will be automatically deterred from proceeding, which has a higher likelihood of encouraging compliance compared to post-violation penalties.

5. **Comprehensive Road Safety:**
    - By addressing both helmet and seatbelt usage, the system covers a broader range of safety gear requirements, contributing to overall road safety, especially in regions with significant motorcycle usage or high rates of traffic-related injuries.

Conclusion

In conclusion, this invention fills a **research gap** by offering a **novel and automated solution** that links real-time AI-based safety gear detection with the control of traffic signals. This unique combination of technologies enhances road safety, provides immediate enforcement, and can be implemented cost-effectively across existing infrastructure.

D. DETAILED DESCRIPTION:

The present invention provides a system and method for ensuring the usage of safety gear—specifically seatbelts and helmets—by vehicle occupants through real-time AI-based detection. The system integrates with existing traffic signal infrastructure to enforce safety gear compliance before allowing vehicles to pass through traffic signals. This system utilizes advanced image recognition, computer vision, and AI algorithms to monitor whether the driver and passengers are wearing their required safety equipment. If they are not, the traffic signal remains red, preventing the vehicle from proceeding.

Overview of the System:

The system consists of several key components that work together to detect seatbelt and helmet usage and control traffic signal operations:

1. **AI-Based Detection System**:
    - The AI-based system uses **high-resolution cameras** installed at traffic signal points to capture images of the vehicle and its occupants.
    - The system processes these images in real-time using **image recognition** algorithms to detect the presence of seatbelts and helmets on the driver and passengers.
    - The system employs **machine learning models** that are trained to recognize seatbelt and helmet usage patterns based on visual features (e.g., straps, helmet shapes, etc.) for both cars and motorcycles.

2. **Traffic Signal Control Mechanism**:
    - Once the AI detection system processes the images, the **control mechanism** communicates with the traffic signal controller to either allow or prevent the signal from changing to green.

    - If the system detects that the driver and/or passengers are not wearing the required safety gear (seatbelts or helmets), the signal will **remain red**. The traffic signal will not allow the vehicle to proceed until all safety gear is properly worn.
    - If the system detects compliance (i.e., the seatbelt is fastened and/or the helmet is worn), the traffic signal will turn **green**, allowing the vehicle to continue its journey.

3. **Camera and Sensor Integration**:
    - The system uses cameras equipped with **high-definition imaging** and possibly **infrared sensors** for low-light conditions, ensuring detection accuracy day or night.
    - The cameras are strategically placed at locations that provide a clear view of the vehicle occupants, such as at intersections or roadside poles, facing the vehicle as it approaches the traffic signal.

4. **Data Processing and Feedback**:
    - The **AI algorithms** process the captured data (images) in real-time. If the required safety gear is detected, a feedback signal is sent to the traffic control system, allowing the signal to turn green.
    - If the system detects non-compliance (e.g., the seatbelt is unfastened or the helmet is missing), the feedback signal will instruct the traffic signal to remain red until the occupant corrects the non-compliance.

5. **User Interaction (Optional)**:
    - In some cases, the system may provide feedback to the driver or passengers. For instance, **visual indicators** or **alerts** on nearby road signs or on the vehicle's dashboard could notify the driver that they are not wearing the necessary safety gear.
    - Additionally, the system could be configured to record incidents of noncompliance for later review by traffic authorities, ensuring that repeat offenders can be identified.

Operation of the System:

1. **Detection Phase**:

    - As a vehicle approaches the intersection, the AI-based detection system activates and begins capturing images of the vehicle and its occupants.

- The system analyzes these images for visual cues that indicate whether the driver and passengers are wearing seatbelts and helmets.
- For motorcycles, the system checks for helmet usage on both the driver and the passenger.

2. **Signal Control Phase**:

    - After detecting whether the required safety gear is worn, the system communicates with the traffic signal controller.
    - If all required gear is in place (seatbelt fastened, helmet worn), the signal controller changes the light to **green**, allowing the vehicle to proceed.
    - If the safety gear is not worn, the signal controller will keep the light **red**, preventing the vehicle from continuing until the necessary gear is worn.

3. **Compliance Check**:

    - The system continuously monitors the vehicle's occupants for any changes in the safety gear status as the vehicle approaches the intersection.
    - If the driver or passenger adjusts their gear (e.g., fastening the seatbelt or putting on the helmet), the system will recheck for compliance and allow the signal to change if all safety gear is properly worn.

4. **Feedback and Notification**:

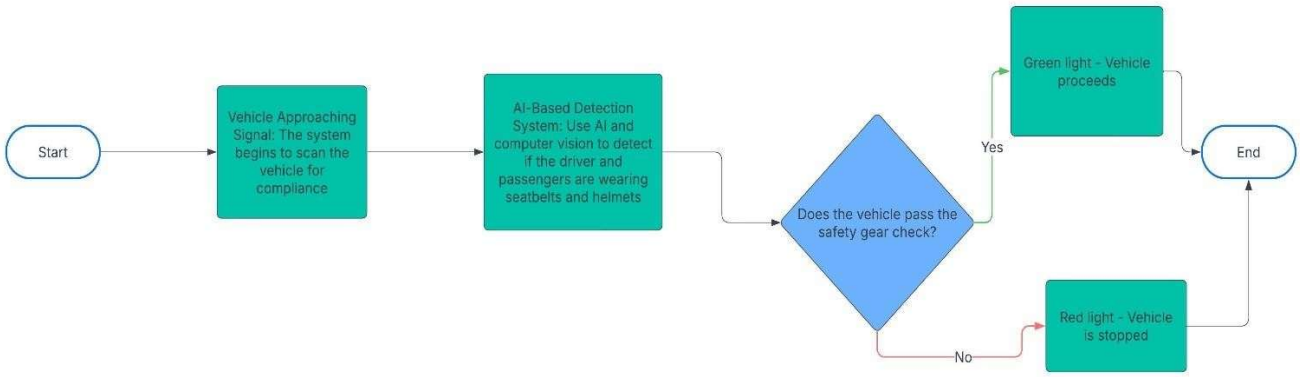
- In the event of non-compliance, the system can trigger visual or audio alerts, notifying the driver and passengers about the need to wear the proper safety gear.
- The system can also log violations, which can be used for further enforcement or follow-up by traffic authorities.

Key Features:

- **Real-Time Detection**: The system operates in real-time, providing instant feedback and control over traffic signals based on the safety gear usage of the vehicle occupants.
- **Dual Safety Gear Detection**: The system uniquely detects both **seatbelt use** in cars and **helmet use** in motorcycles, ensuring that all vehicles comply with necessary safety regulations.
- **Seamless Integration**: The system can be easily integrated with existing traffic control systems, making it cost-effective and scalable for widespread deployment.
- **Cost-Effective and Scalable**: The system uses common traffic cameras and AI algorithms, reducing the need for expensive infrastructure changes while being adaptable for use in various urban or rural environments.

- **User Feedback**: The system can notify the driver and passengers in case of noncompliance, ensuring they are aware of the safety gear requirements.

Process Workflow:



E. RESULTS AND ADVANTAGES:

The AI-Based Safety Gear Detection and Traffic Signal Control System provides numerous **benefits** and **improvements** to road safety, enforcement efficiency, and user experience. The results of implementing this system are substantial, both in terms of **effectiveness** and **practicality**.

1. Increased Road Safety

- **Prevention of accidents**: By ensuring that drivers and passengers wear seatbelts and helmets before proceeding, the system **significantly reduces the risk of injury or death** in the event of an accident. This proactive enforcement encourages consistent safety gear usage, which is a major factor in reducing road-related fatalities.

- **Broader safety coverage**: Unlike traditional systems that might focus only on seatbelt use in cars or helmets for motorcycles, this system ensures that **both seatbelt and helmet compliance** are met for all types of vehicles, enhancing overall safety.

2. Real-Time Compliance Enforcement

- **Instant deterrent**: The system provides a **real-time deterrent** by preventing vehicles from proceeding through traffic signals until the required safety gear is worn. This immediate enforcement helps reinforce road safety rules, unlike current systems that only issue penalties after violations occur.

- **No delays**: Traditional traffic law enforcement can take time to issue fines or warnings. This system offers **instantaneous detection and enforcement**, resulting in a more responsive and efficient method of ensuring compliance.

3. Enhanced Traffic Signal Efficiency

- **Automatic traffic light control**: The system intelligently integrates with traffic signal controllers, adjusting the light based on **real-time detection** of safety gear usage. This reduces the need for manual intervention or reliance on law enforcement, making traffic management more efficient and automated.

- **Fewer traffic violations**: With vehicles being unable to move forward without wearing safety gear, the likelihood of traffic violations (related to seatbelt and helmet use) is **significantly reduced**, leading to smoother traffic flow and better overall safety on the roads.

4. Increased Compliance Rates

- **Preventive enforcement**: The system acts as a proactive tool for compliance rather than just a punitive one. Since vehicles are **prevented from proceeding** until all safety gear is in place, there is a **higher likelihood of encouraging drivers and passengers to wear their safety gear consistently**.

- **Consistency in enforcement**: By automatically monitoring compliance, the system ensures **consistent application** of the rules across all vehicles, with no room for manual discretion or oversight, improving fairness and compliance.

5. Cost-Effective and Scalable

- **Easy integration**: The system is designed to be easily integrated into existing traffic infrastructure, making it **cost-effective** to deploy, especially in cities or regions that already have a robust traffic control system.

- **Scalability**: This solution can be scaled across a variety of locations, including urban and rural areas. Whether implemented in high-traffic cities or on highways, it offers a flexible, **scalable solution** that doesn't require extensive infrastructure changes.

6. Reduced Law Enforcement Workload

- **Less reliance on manual enforcement**: By automating the enforcement of seatbelt and helmet regulations, the system **reduces the need for manual police patrols** or law enforcement presence at intersections for this specific violation. Law enforcement officers can focus on other critical tasks, improving overall police efficiency.

- **Fewer penalties and fines**: By addressing compliance before violations happen, the system minimizes the need for issuing **fines or penalties**, making it less burdensome on both drivers and enforcement officers.

7. Improved Public Awareness and Behavior

- **Raising awareness**: As the system directly informs drivers of their non-compliance, it acts as a **reminder** and educational tool. This is likely to increase public awareness of safety gear usage, which, in turn, can contribute to a cultural shift in road safety behavior.

- **Increased public acceptance**: As people become accustomed to the idea of traffic signals controlling the flow based on safety gear compliance, there is likely to be a positive shift in societal attitudes toward **wearing seatbelts and helmets**.

8. Enhanced Data Collection for Traffic Management

- **Incident tracking and analysis**: The system can collect data on safety gear compliance rates, which can be valuable for authorities to analyze patterns in road safety behavior. This data can be used to improve traffic management strategies, enhance safety campaigns, or identify areas with lower compliance.

- **Long-term improvements**: Continuous monitoring of safety gear compliance provides valuable insights that can help traffic authorities **make data-driven decisions** to improve overall road safety in the long term.

F. EXPANSION: Necessary Variables for the Invention

To ensure that the AI-Based Safety Gear Detection and Traffic Signal Control System functions effectively, several key variables must be considered and integrated into the design and operation of the system. These variables are essential for optimizing the performance, scalability, and adaptability of the system.

1. Vehicle Type Detection

- **V**ariable: Vehicle type (Car, Motorcycle, Bus, etc.)
- Explanation: The system must differentiate between various types of vehicles (e.g., cars, motorcycles, buses, trucks) to ensure it properly checks seatbelt usage for cars and helmets for motorcycles. AI algorithms need to identify the vehicle type and apply appropriate checks for each type.
- Impact on Invention: Ensures that seatbelt checks apply to cars and helmet checks apply to motorcycles. This distinction is crucial to prevent false readings and optimize the detection process.

2. Occupant Detection

- Variable: Number of occupants (driver, passengers)
- Explanation: The system must determine how many occupants are in the vehicle and ensure each one wears the required safety gear. For example, in a car, all passengers must wear seatbelts, and in a motorcycle, both the driver and passenger must wear helmets.

- Impact on Invention: If the vehicle has multiple occupants, the system must verify that each of them is wearing their respective safety gear. This makes the detection process more complex but crucial for accurate enforcement.

3. Seatbelt Detection Technology

- Variable: Seatbelt status (fastened or unfastened)

- Explanation: The system needs to detect whether the seatbelt is fastened or unfastened, which can be done using computer vision or sensor-based technology. The AI needs to recognize the visual patterns of a seatbelt to check if it's in use.
- Impact on Invention: The AI must differentiate between the visual cues of a fastened and unfastened seatbelt, ensuring that vehicles with non-compliant occupants remain stationary at red lights.

4. **Helmet Detection Technology**
   - Variable: Helmet usage (on or off)
   - Explanation: Similar to seatbelt detection, the system needs to recognize if a helmet is being worn by the driver or passenger on motorcycles. Computer vision is used to identify helmets based on their shape, size, and position relative to the person.
   - Impact on Invention: Accurate helmet detection is essential for motorcycles. If the system fails to detect helmet usage, it could cause unnecessary delays for compliant vehicles or fail to stop non-compliant vehicles.

5. **Environmental Factors**
   - Variable: Lighting conditions (day or night)
   - Explanation: The system must function under varying lighting conditions (day, night, fog, or rain). For example, at night, infrared sensors or night-vision cameras might be required to ensure accurate detection, especially for helmet and seatbelt visibility.
   - Impact on Invention: The system must adapt its detection algorithms to account for environmental factors that might affect the clarity of images or video streams.

6. **Traffic Flow and Signal Timing**
   - Variable: Traffic signal timing
   - Explanation: The timing of the traffic light (green, yellow, red) plays a significant role in the system's operations. The AI must determine when to prevent or allow a vehicle to pass through based on whether safety gear is worn.
   - Impact on Invention: The system needs to be integrated with existing traffic control mechanisms, ensuring that safety gear compliance is a factor in real-time traffic light control without causing unnecessary delays.

7. **Real-Time Data Processing**
   - Variable: Processing speed (Real-time detection and response)
   - Explanation: The system must process data in real-time to prevent delays in traffic signal control. The AI algorithms must analyze the images or sensor data quickly enough to make instantaneous decisions about whether to allow a vehicle to proceed.

- Impact on Invention: Slow data processing could lead to delays in signal changes, creating traffic congestion or frustration for drivers. Optimizing the data processing speed ensures smooth operation.

8. **System Scalability**
   - Variable: System integration with existing traffic infrastructure
   - Explanation: The system should be designed to integrate seamlessly with existing traffic signal infrastructure. This means it must be able to function alongside current traffic management systems without requiring a complete overhaul.
   - Impact on Invention: The system's ability to scale and integrate with existing infrastructure makes it easier and more affordable for cities to implement. The system must be adaptable to different traffic management setups, such as urban and rural road networks.

G. WORKING PROTOTYPE/ FORMULATION/ DESIGN/COMPOSITION:

Working prototype is not ready. It will take at least a year to complete it.

H. EXISTING DATA:

While there may not be direct clinical data specifically focused on an AI-based system that combines traffic signal control and safety gear detection, there is existing data and research on related fields that support the effectiveness of such a system. These fields include traffic safety data, seatbelt and helmet compliance, AI-based detection systems, and the impact of enforcement technologies on road safety. Below are some relevant comparative and clinical data points that can help demonstrate the potential of the proposed invention:

1. **Impact of Seatbelt and Helmet Usage on Road Safety**
   - **Seatbelt Usage and Road Safety:**
     - According to the World Health Organization (WHO), seatbelts reduce the risk of death in a crash by 45%–50% for front-seat occupants and 25%–30% for rearseat occupants.
     - Studies show that mandatory seatbelt laws have contributed to significant reductions in road fatalities in countries that enforce these laws strictly.
     - U.S. National Highway Traffic Safety Administration (NHTSA) reports that seatbelt use is one of the most effective methods to reduce injuries and fatalities. In 2020 alone, seatbelt use saved more than 14,000 lives in the U.S.

   - **Helmet Usage and Motorcycle Safety:**
     - According to WHO, helmets can reduce the risk of fatal injury by 40% and the risk of non-fatal injury by 70% in motorcycle accidents.

- A study published by the Insurance Institute for Highway Safety (IIHS) found that helmet laws reduce fatal motorcycle crash rates by 22% on average. In countries with strict helmet laws, motorcycle fatalities are considerably lower than in regions where helmet laws are less enforced.

2. The Effectiveness of Automated Detection Systems in Traffic Safety

- **AI in Traffic Law Enforcement:**

  - Computer vision-based AI systems have been used successfully in traffic camera enforcement (e.g., for detecting speeding or running red lights). Studies have shown that systems like these result in reduced traffic violations and better enforcement efficiency.
  - A study conducted by the University of California found that the use of automated red-light cameras in several cities reduced traffic violations by up to 40% in the first year of implementation. • **AI-Based Seatbelt and Helmet Detection:**

  - Research by companies and universities has demonstrated that AI-powered computer vision can successfully detect seatbelt usage (e.g., AI in vehicle safety systems for seatbelt alerts).
  - The University of Southern California (USC) and NVIDIA have developed AIbased systems that use camera networks to detect seatbelt usage in vehicles in real-time. Their research shows that such systems are highly accurate and provide real-time alerts to the driver, much like how a traffic signal system would function in the proposed invention.

  - Similarly, helmet detection technology has been tested in real-time using computer vision, where systems can detect the presence of helmets for motorcyclists using surveillance cameras or drones.

3. Effectiveness of Traffic Signal Control Systems and Automated Enforcement

- **Red-Light Cameras and Automated Traffic Enforcement:**

  - Studies have found that automated traffic signal systems, such as red-light cameras, reduce traffic accidents at intersections by as much as 21% in hightraffic areas (as seen in cities like New York City, Los Angeles, and London).
  - In regions where automated enforcement is implemented, there has been a marked reduction in the number of violations, including speeding, running red lights, and other traffic infractions, suggesting that real-time enforcement improves overall compliance.

- **Integration of Traffic Signals with Vehicle Monitoring:**

  - Research on integrating traffic signal systems with real-time vehicle monitoring has proven effective in other contexts, such as eco-friendly traffic control (e.g., adjusting signals to prioritize electric vehicles). This shows that traffic signals can be effectively controlled based on vehicle characteristics, including compliance with safety gear requirements.

4. **Data on Public Safety Gear Compliance and Enforcement**

   - **Seatbelt Usage Studies:**

     - In regions with strict seatbelt enforcement, the compliance rates are significantly higher. For example, in countries with mandatory seatbelt laws and high fines, compliance rates can reach over 90%. However, in areas with lower enforcement, compliance rates often drop to 50-70%. The introduction of an automated enforcement system (like the proposed invention) could help bridge this compliance gap by enforcing usage at the moment when it is most necessary. • **Helmet Compliance:**

     - In countries with stringent helmet laws, such as Vietnam and India, helmet use has been shown to increase from 30% to 90% following the introduction of helmet law enforcement programs. Similar studies have shown a reduction in head injuries and fatalities where enforcement is automated and consistent.

5. **Public Response to Automated Enforcement Systems**

   - **Public Perception:**

     - Research has shown that public acceptance of automated enforcement systems (such as speed cameras or red-light cameras) increases when these systems are framed as public safety measures rather than as a revenue-generating tool.
     - In a study by the University of Minnesota, public attitudes toward automated enforcement systems improved when drivers saw them as an efficient tool for enhancing road safety rather than as a means of punishing individuals.

   - **User Compliance:**

     - Studies suggest that the visibility of enforcement (such as cameras and sensors) increases the likelihood that drivers will adhere to safety laws. When a vehicle is unable to move until safety gear is worn, this real-time consequence provides a stronger deterrent than traditional post-violation penalties like fines.

Conclusion on Existing Data:

- Comparative data from existing studies on seatbelt and helmet usage, AI-powered traffic monitoring, and automated law enforcement suggest that the proposed AI-based safety gear

detection system could have a significant impact on road safety by increasing compliance with seatbelt and helmet laws.

- AI and computer vision technologies have already been proven effective in detecting violations and enhancing public safety. When integrated with traffic signal control, this system could provide an immediate deterrent against non-compliance and contribute to lower traffic fatalities and injuries.

4. USE AND DISCLOSURE (IMPORTANT): Please answer the following questions:

A. Have you described or shown your invention/ design to anyone or in any conference?		NO (No)
B. Have you made any attempts to commercialize your invention (for example, have you approached any companies about purchasing or manufacturing your invention)?		NO (No)
C. Has your invention been described in any printed publication, or any other form of media, such as the Internet?		NO (No)
D. Do you have any collaboration with any other institute or organization on the same? Provide name and other details.		NO (No)
E. Name of Regulatory body or any other approvals if required.		NO (No)

5. Provide links and dates for such actions if the information has been made public (Google, research papers, YouTube videos, etc.) before sharing with us. **NA**

6. Provide the terms and conditions of the MOU also if the work is done in collaboration within or outside university (Any Industry, other Universities, or any other entity). **NA** 7. Potential Chances of Commercialization. **Yes**

8. List of companies which can be contacted for commercialization along with the website link.

Here are two companies that specialize in electric vehicle charging solutions and could be potential partners for the commercialization of the Self-Sufficient Charging System for Electric Public Transportation: **1. Iteris**

- Description: Iteris is a leading provider of smart transportation systems, including AI-powered traffic control and safety solutions. They focus on traffic management, road safety, and AI-driven analytics.
- **Website: https://www.iteris.com**

2. Kapsch TrafficCom

- Description: Kapsch specializes in intelligent traffic systems (ITS) that include solutions for traffic enforcement, smart mobility, and vehicle detection.
- **Website: https://www.kapsch.net**

9. Any basic patent which has been used and we need to pay royalty to them.

10. **FILING OPTIONS:** Please indicate the level of your work which can be considered for provisional/ complete/ PCT filings - (Provisional)

11. **KEYWORDS:**

- **AI-Based Traffic Control**
- **AI Safety Gear Detection**
- **Automated Traffic Signal Control**
- **Real-Time Traffic Enforcement**
- **Seatbelt Detection System**
- **Helmet Detection System**
- **Traffic Signal Automation**
- **Safety Gear Compliance System**
- **Computer Vision Traffic Enforcement**
- **Vehicle Safety Gear Monitoring**
- **Smart Traffic Signal System**
- **Real-Time Compliance Monitoring**
- **Vehicle Safety Technology**
- **Automated Road Safety Enforcement**
- **AI Traffic Violation Detection**
- **Real-Time Road Safety Solutions**
- **Traffic Control with AI Integration**
- **Automated Seatbelt Detection**
- **Helmet Law Enforcement System**
- **Vehicle Monitoring System**
- **Traffic Safety Gear Detection**
- **AI-Powered Traffic Management**
- **Traffic Signal Enforcement System**
- **Road Safety AI System**

- **AI-Driven Compliance Enforcement**